Chapter 7

Morphology of Bacteria

Chapter Outline

7.1 Bacterial Size, Shape and Arrangement

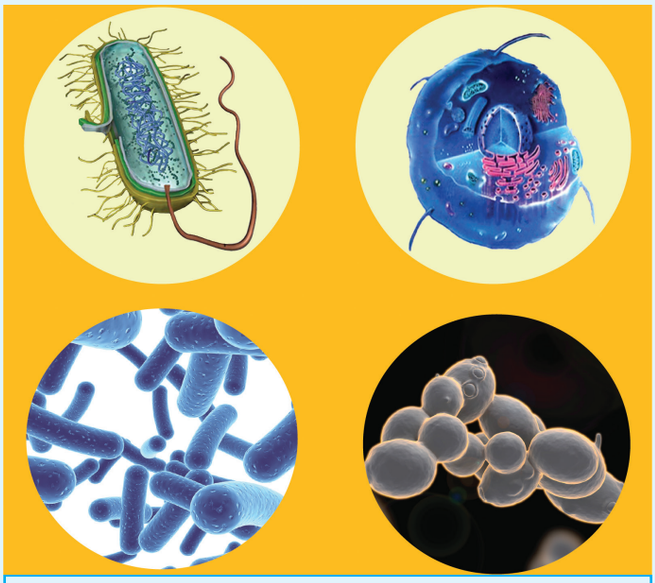
7.2 Structures External to Cell Wall of Bacteria

7.3 Cell Envelope of Bacteria

7.4 Structures Internal to Cell Membrane of Bacteria

7.5 Eukaryotic Cell Structure

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The distinction between **prokaryotes** and **eukaryotes** is considered to be the most important distinction among groups of organisms. Eukaryotic cells contain membrane bound organelles, such as mitochondria, while prokaryotic cells do not.

Learning Objectives

After studying this chapter the student will be able,

- *To know the size, shape and arrangement of bacteria.*
- *To list a few examples of bacteria with their shapes.*
- *To understand and describe the role of the structures external to the cell wall.*
- *To understand the structure, function and arrangement of bacterial flagella.*
- *To describe the role of capsule, slime layer, pili, flagella and fimbriae in a prokaryotic cell.*
- *To describe the structure and function of cell wall, outer membrane and cell membrane.*
- *To know the significance of Cell Envelope.*
- *To differentiate between Gram positive and Gram negative bacteria.*
- *To know the structures and functions internal to cell membrane.*
- *To differentiate between prokaryotic and eukaryotic cell structure.*

Living organisms are differentiated from non living matter by their (1) ability to reproduce (2) ability to ingest or assimilate food and metabolize them for energy and growth (3) ability to excrete waste products (4) ability to react to changes in their environment (irritability) and (5) susceptibility to mutation. The living organisms include a variety of micro and macro organisms of different size, shape, morphology and behaviour. They include tiny bacteria, protozoans, worms, plants and animals.